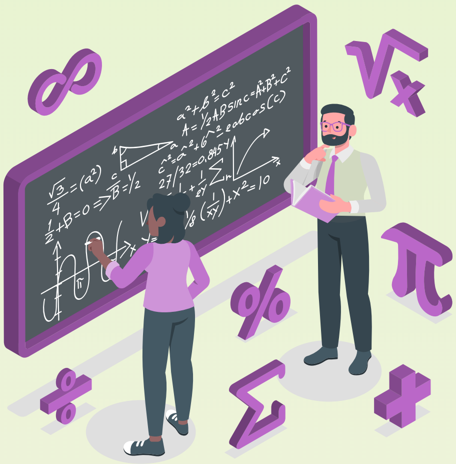

LINEAR EQUATIONS IN TWO VARIABLES

Linear Equation

- An equation in which the power of variable is one, is called a linear equation.

For eg.

$4x + 5 = 3x + 1$

$2x + 3y = 45$

Solution of a linear equation in one variable

- Value of the variable which when substituted in equation, makes two sides of the equation equal, is called the solution of the equation.
- For e.g. Consider the equation $2x + 7 = 0$. Its solution i.e. the root of the equation is $-7/2$. This can be represented on the number line.

Linear equations in two variables

- An equation which can be expressed in the form $ax + by + c = 0$, where a, b and c are real numbers such that a and b are not zero, is called a linear equation in two variables x and y.
- In the linear equation $ax + by + c = 0$, a is coefficient of x, b is coefficient of y and c is constant. Here a, b, c are called the coefficients of the linear equation.

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Solution of linear equations in two variables

Solution of linear equation in two variables i.e. $ax + by + c = 0$ has a pair of values, one for the unknown variable x and one for the second unknown variable y which satisfy the given equation.

For eg.

$2x + 3y = 6$. When we substitute $x = 1$, $y = 4/3$ in the given equation, we find that $2 \times 1 + 4/3 \times 3 = 2 + 4 = 6$. Therefore $(1, 4/3)$ is a solution of the linear equation.

Graph of a linear equation in two variables

In order to draw the graph of a linear equation $ax + by + c = 0$, $a \neq 0$, $b \neq 0$, we may follow the following algorithm.

Step-I : Obtain the linear equation and let the axis equation be $ax + by + c = 0$.

Step-II : Express y in terms of x to obtain $y = -\left(\dfrac{ax+c}{b}\right)$

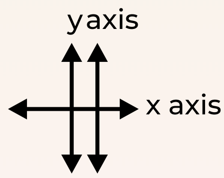
Step-III : Put any two or three values of x and calculate the corresponding values of y from the expression in step-II to obtain two solution say (a_1, b_1) and (a_2, b_2), if possible take values of x as integers in such a manner that the corresponding values of y are also integers.

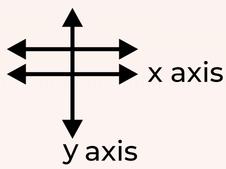
Step-IV : Plot points (a_1, b_1) and (a_2, b_2).

Step-V : Join the points marked in step IV to obtain a line. The line obtained is the graph of the equation $ax + by + c = 0$.

Equations of Lines Parallel to the x-axis and y-axis

$x = a$ is an equation of line parallel to y-axis and $y = b$ is an equation of line parallel to x – axis.





For eg.

Represent $x - 5 = 0$ on the Cartesian plane.

Sol. We solve $x - 5 = 0$; $x = 5$; Here, the graph AB is line parallel to the y-axis and at a distance of 5 units to the right of it.